Conventional pansharpening

Launcher: `Toolbox/pansharpening.bat`

Purpose

Pansharpening transfers the finer spatial detail of a single-band panchromatic (PAN) image to a lower-resolution multispectral (MUL) image. It can make small crop-mark patterns easier to see, but it creates a derived image and can introduce spectral or edge artefacts.

Inputs and parameters

Option	What to provide	Behaviour
Input MUL Image	Corrected multiband GeoTIFF	The image is aligned to the PAN grid internally. Use the matching acquisition and footprint.
Input PAN Image	Corrected single-band panchromatic GeoTIFF	Provides the output grid and spatial detail. It should share the MUL image's CRS and acquisition.
Output Folder	Folder, or blank	Blank writes beside the MUL image. A missing entered folder is created only at its final level.
Vector layer	Optional clip polygon	Crops the final pansharpened results (including the WV3 NIR2 product) and assigns NoData `−9999`. The unclipped temporary results are removed. Inputs that were already clipped during atmospheric correction are also supported: their NoData is kept out of the calculation.

Option	What to provide	Behaviour
Sensor	`WV3` or `LEGION`	Used by weighted Brovey. Default: `WV3`. Sensor choice does not affect Bayesian processing.
Algorithm	`Bayesian` or `wBrovey`	Selects the fusion method. Default: `Bayesian`.
lambda	Floating-point value from 0 to 1	Bayesian regularization/weighting parameter. Default: `0.995`. The field is validated even when `wBrovey` is selected.
Output compression	`Lossless` or `Near-lossless`	Lossless (default) is bit-exact. Near-lossless (LERC) allows an error of at most 5e-5 per pixel and only applies to Float32 outputs; when both inputs hold integers, the output is an integer and always lossless. Keep the output lossless if it feeds PCA, spectral indices or MAE: see Output compression.

Algorithms

Bayesian uses Orfeo ToolBox's Bayesian method. The MUL raster is aligned with nearest-neighbour resampling. The method uses whole-image statistics, so on inputs with NoData the values can differ by a fraction of a percent from a run on unclipped inputs. Higher lambda values impose a stronger PAN-driven constraint. The default is intended for reflectance scaled to integer-like values; with unscaled 0–1 reflectance, it can overwhelm the spectral signal and produce nearly identical bands or extreme values.

wBrovey injects PAN detail using sensor-specific spectral weights and linear MUL resampling. WV3 uses seven weighted bands, then pansharpens NIR2 separately with local mean/variance matching and creates a VRT stack. The current LEGION sensor has no Brovey weights configured, so `wBrovey` is not operational for LEGION; use Bayesian instead.

Outputs

- Bayesian: `<MUL-name>_bayes.tif`
- Weighted Brovey: `<MUL-name>_brovey.tif`
- WV3 weighted Brovey also creates an NIR2 LMVM product and a `HIGHRES_stack.vrt` that combines the products.

Pixels where an input has NoData are NoData in the outputs (`-9999`, or `0` for unsigned integer outputs).

NIR2 edge margin

The NIR2 LMVM product is also NoData within 3 pixels of an input NoData pixel. LMVM computes each output pixel from the mean and variance of the PAN and NIR2 values in a window of 3 pixels around it. Next to a NoData area, that window includes pixels that hold no real data, so the local statistics are wrong and single pixels can be off by 3–5 % of their value. Weighted Brovey works pixel by pixel and has no such window, so bands 1–7 keep their full extent.

The margin only appears when the inputs already contain NoData, for example when the images were clipped during atmospheric correction. Band 8 of `HIGHRES_stack.vrt` then ends 3 pixels (about 1 m for WV3) inside the edge of bands 1–7; use band 8 as the mask when all eight bands are needed. Inputs without NoData are not affected, and neither is a clip made with this tool's **Vector layer** option, because that clip runs after the fusion and gives all eight bands the same extent.

Main outputs are Cloud Optimized GeoTIFFs: DEFLATE (lossless), or LERC (near-lossless, Float32 only). Intermediate alignment and calculation rasters are deleted after successful processing.

Recommended settings and checks

- For DOS3 outputs scaled by `10000`, begin with Bayesian lambda `0.995`.
- For reflectance in the 0-1 range, begin near `0.1`, then compare results rather than assuming one value is universally correct.
- Check that output bands have plausible and distinct min/max, mean, and standard deviation values. Nearly identical statistics in every band indicate spectral collapse.
- Compare sharp linear features with the original PAN. Doubled edges, halos, and shifts may indicate input misregistration rather than archaeological structure.
- Keep the unsharpened multispectral data for quantitative spectral analysis; pansharpened pixels are modelled values.